AIRWALLEX DEVELOPER HACKATHON

FlowGuard

Adaptive treasury control and AP duplicate protection

A safer payment path for cross-border small businesses

Reserve-aware decisions. Deterministic duplicate detection. Intent-bound approvals.

PAYMENT GUARDRAILS

$15,000

protected reserve floor

DUPLICATE INVOICE

BLOCKED

INTENT TOKEN

Verified before payout

Liquidity risk sits at the intersection of timing, fraud, and opaque cost

Small cross-border firms need more than a payment queue. They need a defensible decision engine.

>60%

of small businesses cite cash-flow volatility and uneven supplier obligations as a major risk

1.29%

average duplicate invoice rate reported for SMBs, often triggered by minor invoice-number variations

1.5%-6%

typical SME FX markup range cited in cross-border payment research

Operational exposure compounds when payments bypass a reserve policy or rely on exact-string invoice matching.

Sources: Federal Reserve Small Business Credit Survey; IOFM and SAP Concur SMB research; BIS CPMI Roadmap

THE SOLUTION

FlowGuard turns payment approval into a policy-checked decision

Each payment plan must preserve liquidity, clear duplicate controls, and carry a verifiable intent record.

01

Protect cash

Maintains a configurable USD reserve floor before funds leave a wallet.

02

Identify duplicates

Detects exact, fuzzy, and proximity matches before payment can proceed.

03

Bind intent

Links approval to a cryptographic plan hash and immutable audit record.

Result: a payment is explainable before it is executable.

TREASURY DECISIONING

The reserve floor determines whether an invoice can move today

The engine calculates a projected base-currency balance before it recommends a payout.

Projected balance

= Current liquidity - (Invoice amount x FX rate + corridor fee)

CURRENT LIQUIDITY

$33,110

RESERVE FLOOR

$15,000

PAYMENT OUTCOME

PAY NOW or DEFER

Scenario A begins with $33,110 across four wallets and a $15,000 reserve floor. Critical invoices can pay, lower-priority obligations defer when the floor is threatened.

AP DUPLICATE DEFENSE

A near-match invoice does not reach the payment rail

FlowGuard combines deterministic checks that conventional exact matching misses.

01

Normalized exact match

Removes punctuation, slashes, and spaces before comparison

02

Fuzzy identifier match

Flags Levenshtein distance <= 2 after re-bill suffix handling

03

Triple-vector proximity

Checks vendor, amount, and a 14-day invoice-date window

SCENARIO B

Bavaria Precision

INV-2024-884

INV-2024-884B

CRITICAL DUPLICATE

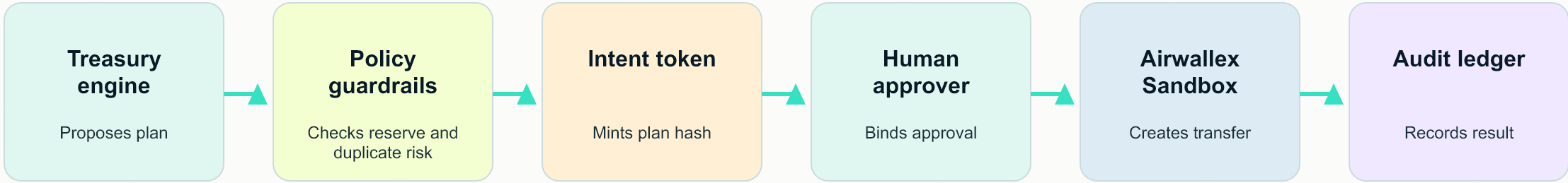
Payment blocked

The approval path ends before an intent token can be minted.

INTENT GOVERNANCE

ArmorIQ binds a payment decision to policy and human approval

The plan has to pass the guardrails before FlowGuard sends a transfer request.



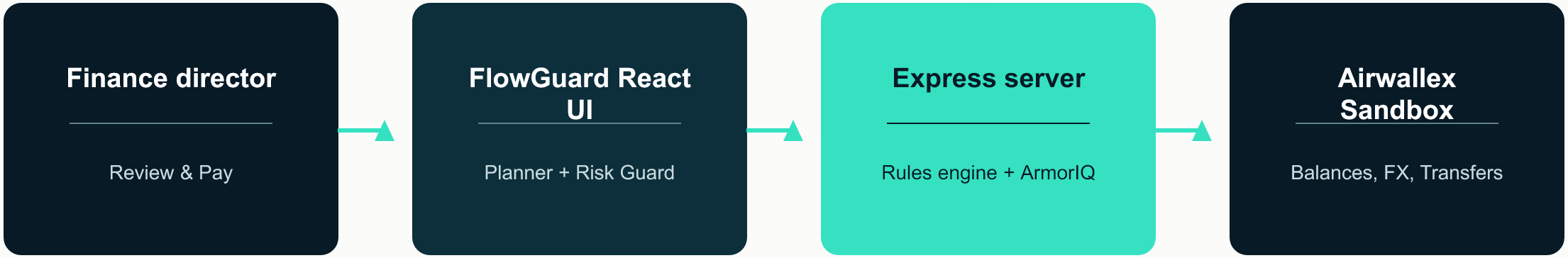
Policy violation: FlowGuard halts execution and records the blocked event.

Guardrails: reserve floor intact, no duplicate risk, payout within ceiling

SANDBOX ARCHITECTURE

A React interface, policy-aware backend, and Airwallex Sandbox work as one payment loop

Live balances and FX inform a plan. A verified approval creates and settles a sandbox transfer.



Selected calls: GET balances/current, GET fx/rates/current, POST transfers/create, POST simulation/transfers/{id}/transition

Authentication uses a short-lived bearer token. Every transfer carries a UUID request_id for idempotency.

DEMO SCENARIOS

Four preset situations make the product behavior visible in three minutes

Each scenario changes the inputs and lets the rules engine show why the recommendation changes.

Scenario	Financial event	FlowGuard response
A Baseline	$33,110 across four wallets; $15,000 reserve floor	Pays critical obligations while preserving buffer
B Duplicate attack	Injects Bavaria invoice INV-2024-884B	Blocks duplicate; stops intent token minting
C Delayed receipt	$14,500 customer receipt moves 14 days	Defers Shenzhen MicroSensors to protect floor
D Deposit replenishment	Adds $25,000 USD via sandbox deposit	Unlocks previously deferred obligations

Use the scenario selector live: show the condition, then show the policy decision.

LIVE DEMO PLAN

A three-minute walkthrough follows the risk from detection to settlement

Keep the browser open with Scenario A active, then change scenarios only when the storyline calls for it.

00:00

Problem and promise

Liquidity volatility, duplicate invoices, and the reserve floor

00:40

Payment planner

Show balances, corridor costs, and PAY NOW versus DEFER

01:20

Duplicate attack

Inject the Bavaria near-match and show the policy block

02:05

Verified transfer

Review intent token, submit sandbox transfer, simulate PAID

02:40

Audit trail

Close on traceability and safe autonomous finance

PRESENTER CUE

Show the reason behind every decision.

Judges should see a specific control, a specific outcome, and an audit record.

CLOSING

FlowGuard gives autonomous payments a policy boundary.

It protects the cash buffer, detects suspicious duplicates, and links each executed transfer to a verified approval path.

Airwallex Sandbox + ArmorIQ Intent Assurance + Deterministic treasury rules

Thank you